



Decision Tree

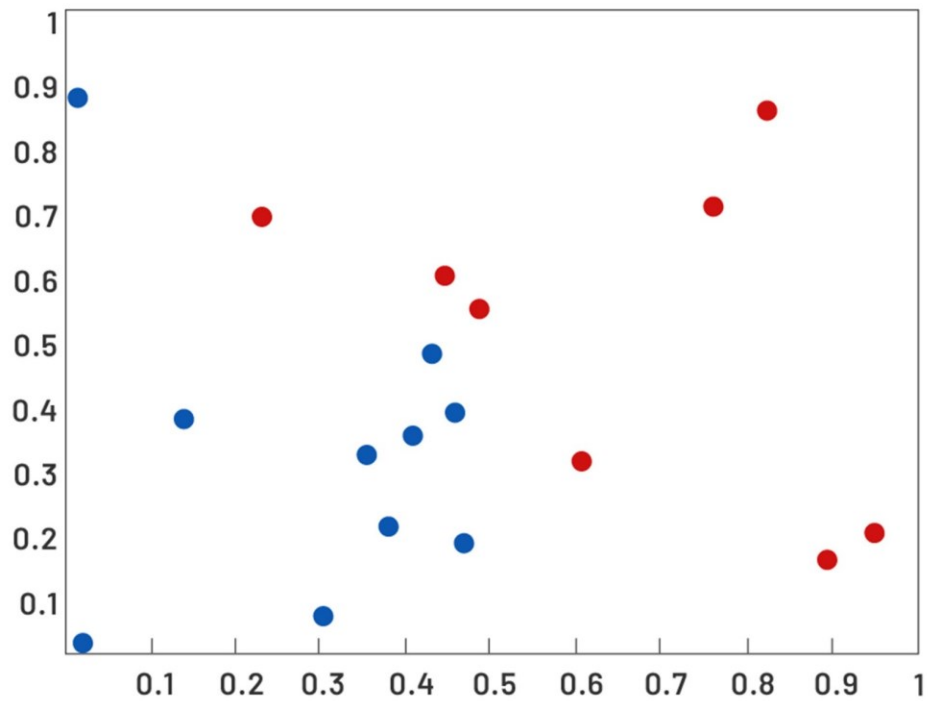
Ehsan Nazerfard

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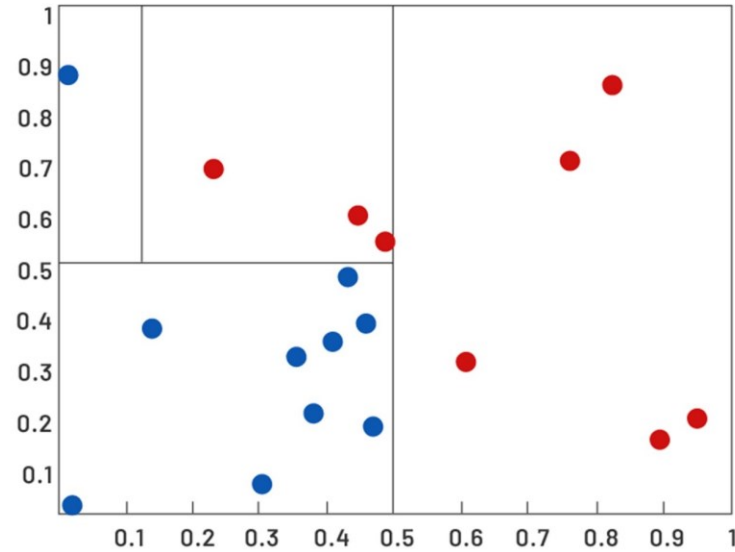
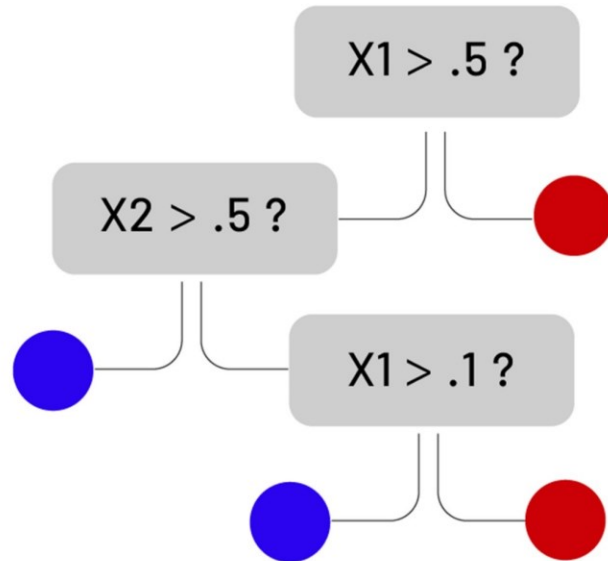
Decision Boundary

► What kind of Decision Boundary for this data?



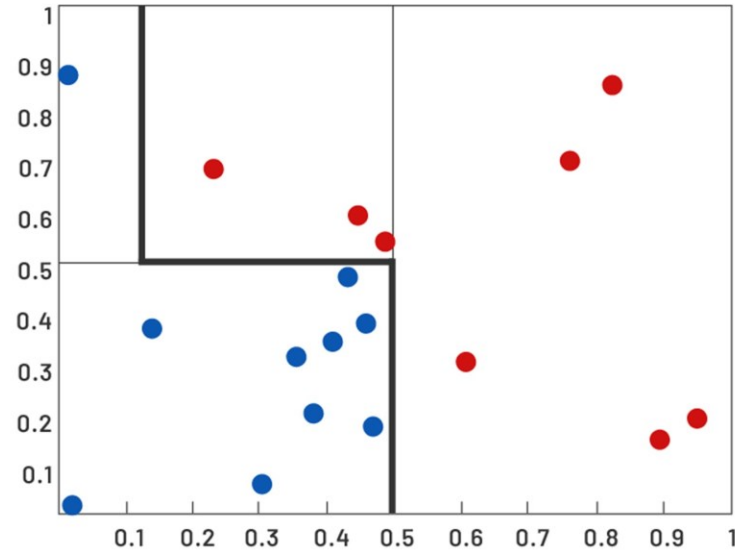
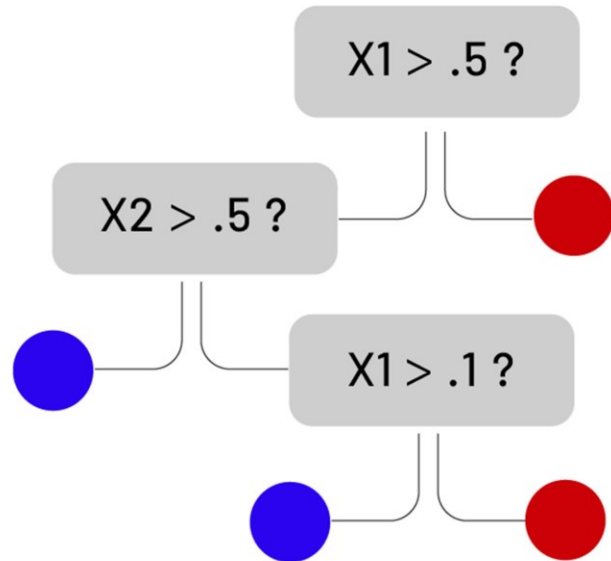


Decision Tree Classifier





Decision Tree Classifier (cont.)



What are the challenges?

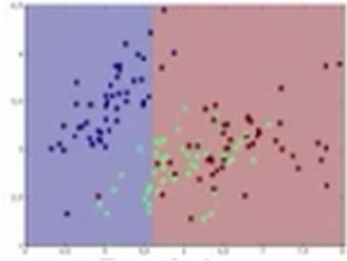


-
- COMplexity
(Tree Depth)
-
- Bias
(underfitting)
- Variance
(overfitting)
- +
- Decision Stump

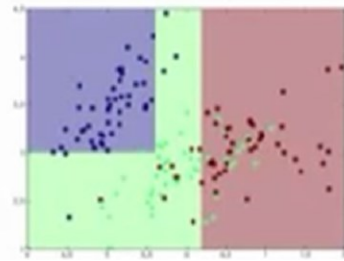


Controlling Complexity

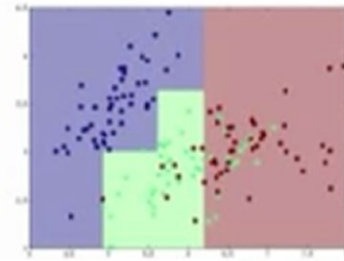
► Maximum depth cutoff



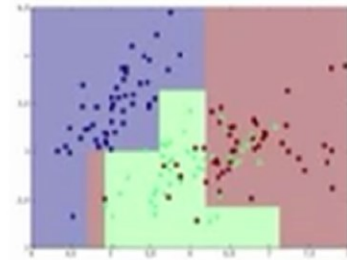
Depth 1



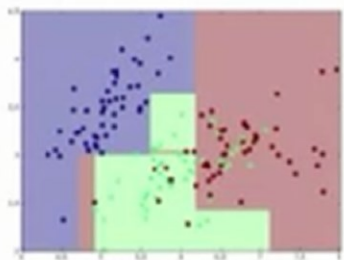
Depth 2



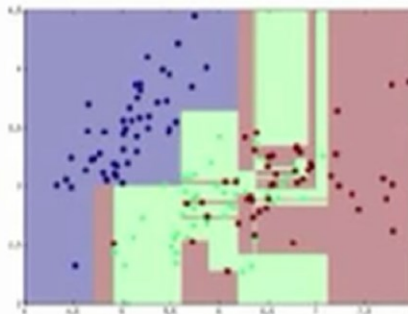
Depth 3



Depth 4



Depth 5



No limit



The Akinator

► What would be the most important question to ask first?





Entropy

- ▶ Entropy → Information Gain
- ▶ Entropy is a measure of randomness
- ▶ Example:
 - ▶ Communicating fair coin tosses:
HTTHTHHHTTHTT ...
 - ▶ Communicating my daily lottery results
000000 ...
 - ▶ The latter, takes less work to communicate. Why?



Entropy (cont.)

► Formal Definition

$$\text{► } H(X) = \sum_i p(x_i) \log_2 \frac{1}{p(x_i)}$$

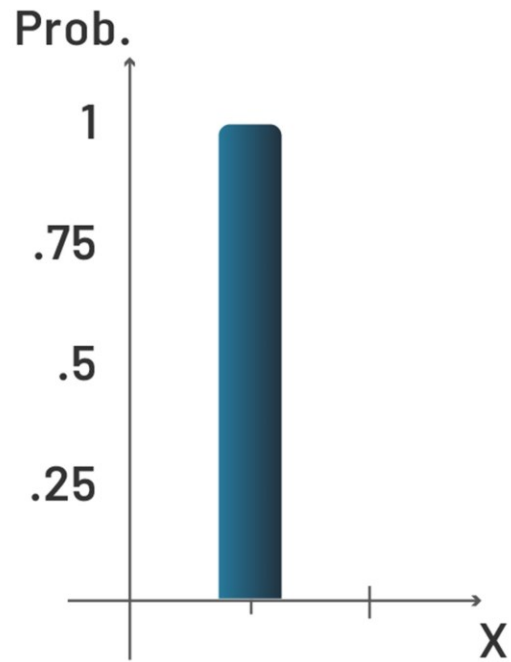
► Shannon got the letter H from Boltzmann's H-theorem

► $E(X)$ measures the randomness of random variable X .

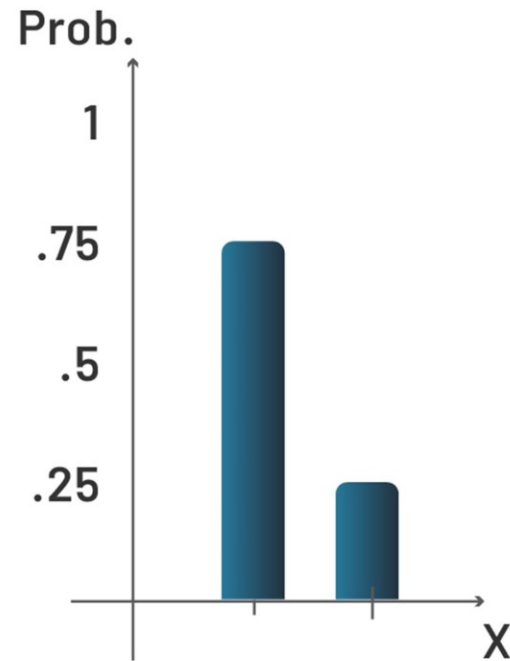
$$E(X) = - \sum_i p(x_i) \log_2 p(x_i)$$



Entropy (cont.)



$$E(X) = -(\log_2 1 + 0 \times \log_2 0) = 0$$



$$E(X) = -(0.75 \log_2 0.75 + 0.25 \log_2 0.25) = 0.81$$



Entropy (cont.)

- ▶ $E(S)$ measures the impurity of set S .
- ▶ Given a collection of S , containing positive and negative examples of some target concept, the entropy of S relative to this boolean classification is:

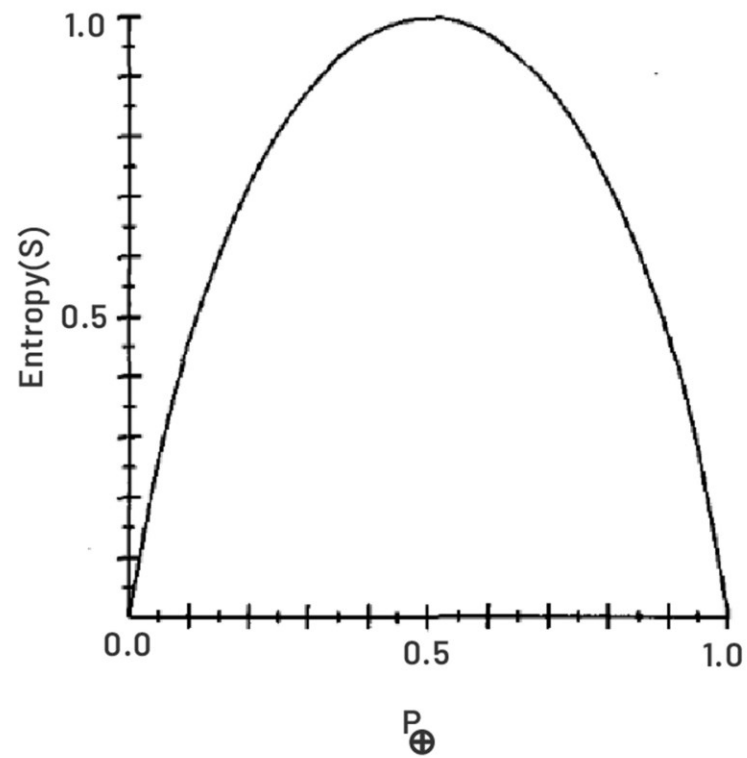
$$E(S) = -(p_{\oplus} \log_2 p_{\oplus} + p_{\ominus} \log_2 p_{\ominus})$$

- ▶ $p_{\oplus}$ refers to the proportion of positive examples.



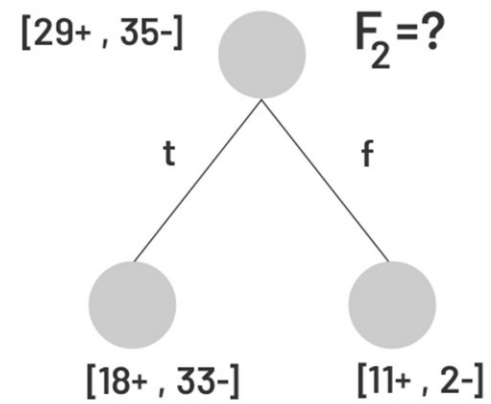
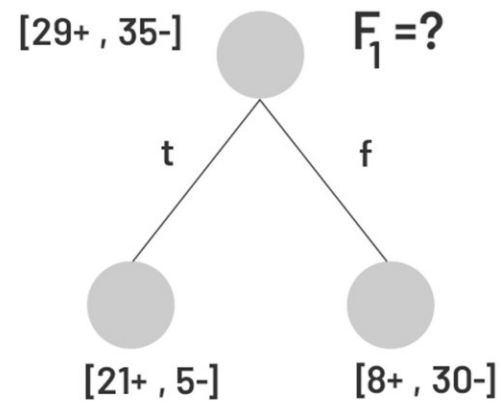
Entropy Function

► The entropy function relative to a boolean classification, as $p_{\oplus}$ varies between 0 and 1.





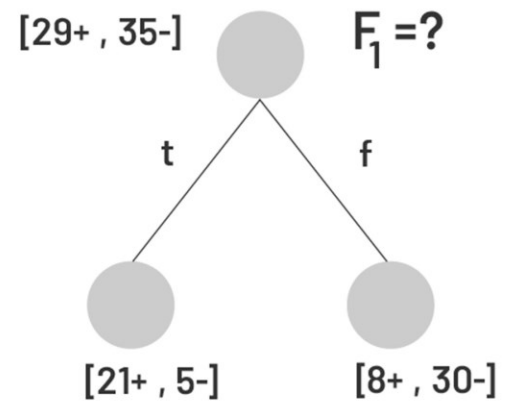
Which Feature is more Effective*?



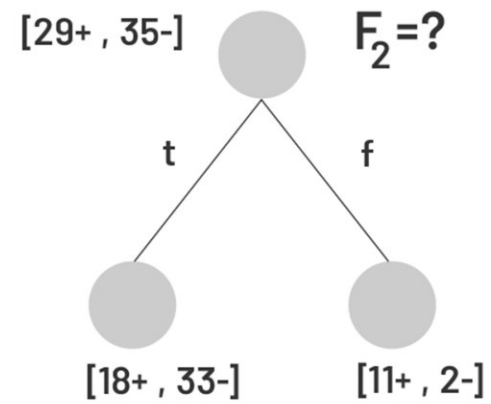
*effective here indicates dichotomiser



How Come?



$$IG([29+ , 35-], F_1) = 0.26$$



$$IG([29+ , 35-], F_2) = 0.12$$

IG (Information Gain)



Information Gain

- ▶ Information Gain measures the expected reduction in entropy.
- ▶ Formally, the information gain, $(\text{Gain } S, F)$ of a feature F , relative to a collection of examples S , is defined as:

$$\text{Gain}(S, F) = \text{Entropy}(S) - \sum_{v \in \text{Values}(F)} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

- ▶ where $\text{Values}(F)$ is the set of all possible values for feature F , and S_v is the subset of S for which feature F has value v .



The PlayTennis/Weather Data

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No



The Wind Feature

Values(Wind) = Weak, Strong

$S = [9+, 5-]$

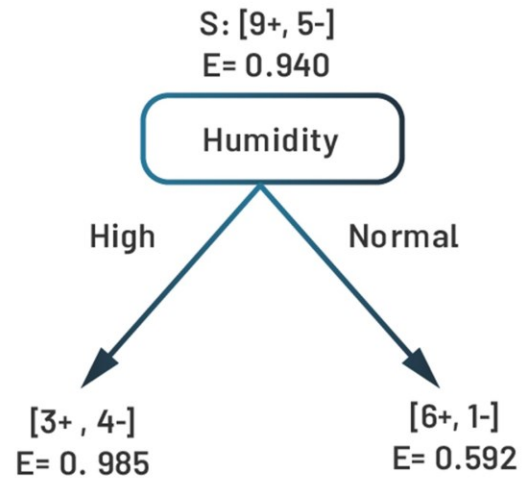
$S_{\text{Weak}} \leftarrow [6+, 2-]$

$S_{\text{Strong}} \leftarrow [3+, 3-]$

$$\begin{aligned}\text{Gain}(S, \text{Wind}) &= \text{Entropy}(S) - \sum_{v \in \{\text{Weak}, \text{Strong}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v) \\ &= \text{Entropy}(S) - (8/14) \text{Entropy}(S_{\text{Weak}}) \\ &\quad - (6/14) \text{Entropy}(S_{\text{Strong}}) \\ &= 0.940 - (8/14)0.811 - (6/14)1.00 \\ &= 0.048\end{aligned}$$

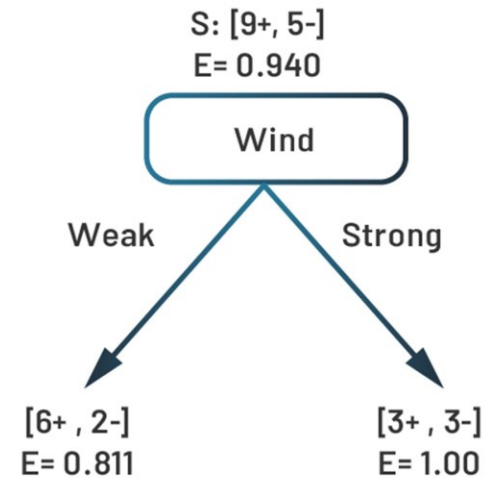


Which Feature is more Effective?



$$\begin{aligned}\text{Gain}(S, \text{Humidity}) &= .940 - (7/14) \cdot .985 - (7/14) \cdot .592 \\ &= .151\end{aligned}$$

$$\text{Gain}(S, \text{Humidity}) = 0.151$$

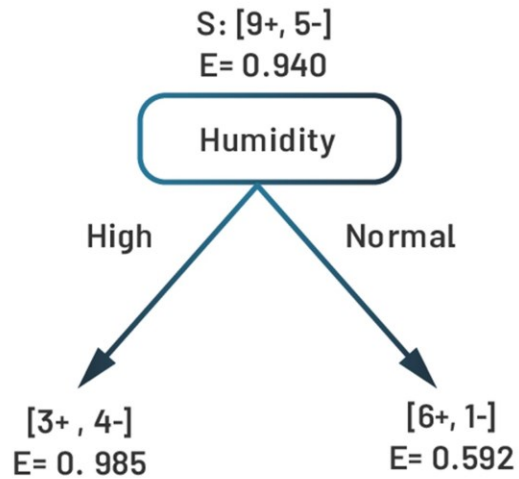


$$\begin{aligned}\text{Gain}(S, \text{Wind}) &= .940 - (8/14) \cdot .811 - (6/14) \cdot 1.0 \\ &= .048\end{aligned}$$

$$\text{Gain}(S, \text{Wind}) = 0.048$$



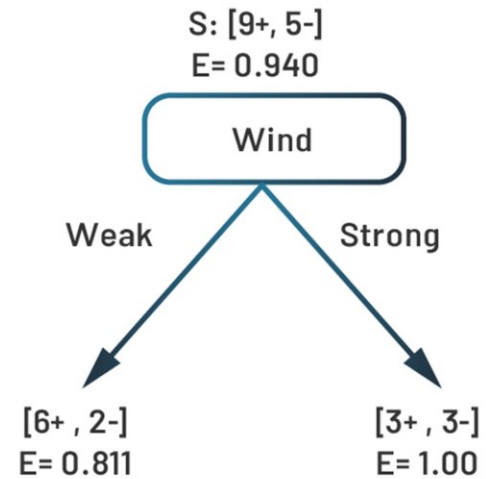
Which Feature gives the highest gain?



$$\begin{aligned}\text{Gain}(S, \text{Humidity}) &= .940 - (7/14) \cdot .985 - (7/14) \cdot .592 \\ &= .151\end{aligned}$$

$$\text{Gain}(S, \text{Humidity}) = 0.151$$

$$\text{Gain}(S, \text{Outlook}) = 0.246$$



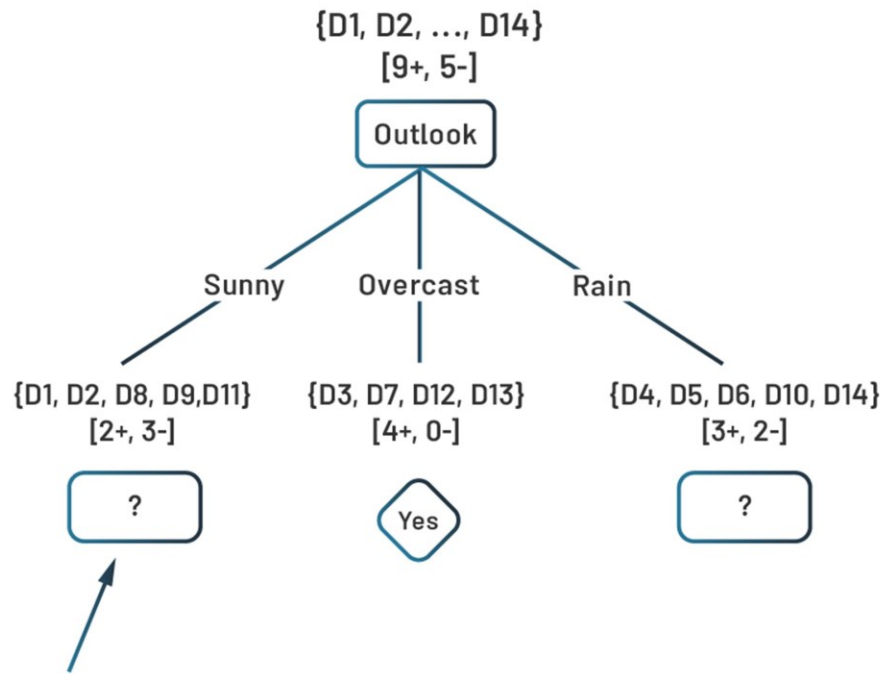
$$\begin{aligned}\text{Gain}(S, \text{Wind}) &= .940 - (8/14) \cdot .811 - (6/14) \cdot 1.0 \\ &= .048\end{aligned}$$

$$\text{Gain}(S, \text{Wind}) = 0.048$$

$$\text{Gain}(S, \text{Temp}) = 0.029$$



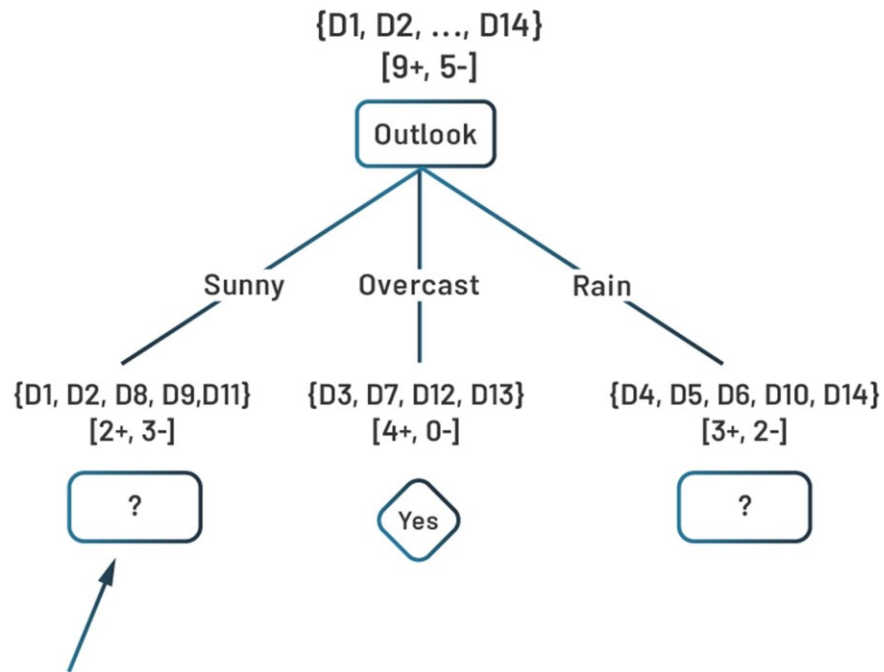
What about the Next Feature



Which attribute should be tested here?



What about the Next Feature



Which attribute should be tested here?

$S_{\text{Sunny}} = \{D1, D2, D8, D9, D11\}$

$\text{Gain}(S_{\text{Sunny}}, \text{Humidity}) = .970 - (3/5) 0.0 - (2/5) 0.0 = .970$

$\text{Gain}(S_{\text{Sunny}}, \text{Temp}) = .970 - (2/5) 0.0 - (2/5) 1.0 - (1/5) 0.0 = .570$

$\text{Gain}(S_{\text{Sunny}}, \text{Wind}) = .970 - (2/5) 1.0 - (3/5) .918 = .019$



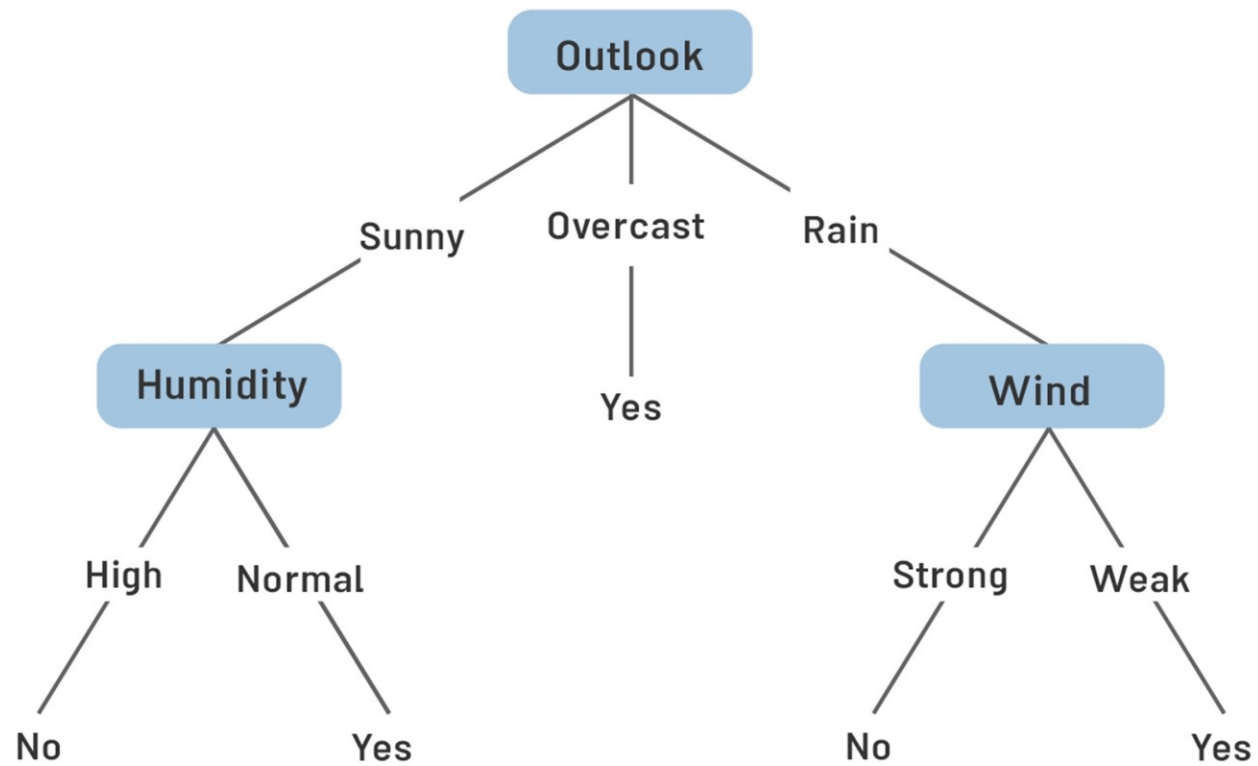


The Weather Data

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
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D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No



The Final Decision Tree





Observations

► Observations?

- Where has the Temperature feature gone?
- Dealing with random noise in the training data:

$D_{15} = (\langle Outlook = Sunny, Temperature = Hot, Humidity = Normal, Wind = Strong \rangle , Play = No)$

- Regarding the Day feature



C4.5 (J48 in Weka)

► C4.5 vs. ID3

	ID3	C4.5
Pruning	x	√
Features	nominal	nominal + numeric
missing values	x	√
Gain	Basic Info Gain	Gain Ratio



The C4.5 Algorithm

► C4.5 vs. ID3

	ID3	C4.5
Pruning	x	√
Features	nominal	nominal + numeric
missing values	x	√
Gain	Basic Info Gain	Gain Ratio

► MDLPC

Temp:	40	48	60	72	80	90
Play:	-	-	+	+	+	-



The C4.5 Algorithm

► C4.5 vs. ID3

	ID3	C4.5
Pruning	x	√
Features	nominal	nominal + numeric
missing values	x	√
Gain	Basic Info Gain	Gain Ratio

$$GainRatio(S, F) = \frac{Gain(S, F)}{SplitEntropy(S, F)}$$

$$SplitEntropy(S, F) = - \sum_{i=1}^{|Values F|} \frac{|S_i|}{|S|} \log \frac{|S_i|}{|S|}$$



Further Reading

- ▶ The CART algorithm
 - ▶ Classification and Regression Tree
- ▶ Gini index
- ▶ Random Forest (later on in this course)



▶ Tom Mitchell, Machine Learning, McGraw-Hill, 1997